

3.1.14**fundamental losses**

losses of a motor running with rated voltage at fundamental frequency (usually 50 Hz or 60 Hz), without additional losses caused by voltage and current harmonics

3.2 Symbols

acc_{coeff}	is the ratio of the maximum torque over rotor inertia, $1/s^2$
f	is the frequency at any operating point, Hz
f_N	is the rated frequency, Hz
I_N	is the rated current, A
J	is the rotor inertia, kgm^2
n	is the speed at any operating point, min^{-1}
n_{max}	is the maximum speed, min^{-1}
n_N	is the rated speed, min^{-1}
n_{90}	is 90 % of the rated speed, min^{-1}
η_n	is the nominal efficiency, %
η_N	is the rated efficiency, %
η_{90}	is the efficiency determined at 90 % speed and full-load, %
η_{ref}	is the reference efficiency according to Tables 2 to 6 and 4.10, %
P_N	is the rated output power, kW
P_{90}	is the output power at 90 % speed and rated torque, kW
r_{HL}	is the additional harmonic loss factor, %
T_N	is the rated torque, Nm
T_{max}	is the maximum torque, Nm
T_{100}	is the full-load torque determined from rated speed and rated power, Nm
U_N	is the rated voltage, V

4 Efficiency classification**4.1 Determination****4.1.1 General**

This clause deals with AC motors that are exclusively rated for variable frequency and voltage operation.

Motors that fall under both IEC 60034-30-1 and IEC 60034-30-2 standards (that are AC motors, which are rated for both on-line (grid) operation and variable speed operation (dual rated motors), for example most induction-motors or line-start permanent-magnet motors), shall bear the IE efficiency class according to the procedures laid out in IEC 60034-30-1 only.

The manufacturer shall use a comparable frequency converter for testing of the motor according to IEC TS 60034-2-3.

4.1.2 Auxiliary devices

Some electric motors covered by this document may be equipped with auxiliary devices such as shaft seals, external fans, mechanical brakes, back-stops and unidirectional bearings, speed sensors, tacho-generators in various combinations.

However, as long as these auxiliary devices are not an integral part of the basic motor design, the determination of efficiency in all possible combinations is not practical. It is recommended that tests for efficiency of such modified standard motors are performed on basic motors with original cooling without auxiliary devices installed.

The power consumption of a separately driven fan is to be included in the efficiency determination procedure when the external fan is an integral part of the basic motor construction. When the external fan is just an optional add-on to a mass-produced motor, which normally carries a shaft-mounted fan, the losses of the basic motor (with the shaft-mounted fan) can be used to determine efficiency.

All vertical motors may be tested horizontally. Angular-contact bearings (thrust bearings) for vertical mounted motors may be replaced by standard bearings during efficiency testing.

Roller bearings may be replaced by standard bearings during efficiency testing.

For all motors where contact seals are accessible from the outside without dismantling of the motor (dismantling of the fan-cover and the fan is accepted), external contact seals should be removed for efficiency testing.

Electro-mechanical brakes shall be removed during testing of motor efficiency. When the motor construction prohibits the removal of the brake, the brake-coil shall be energized from a separate power source and the energy consumption of the brake-coil shall be disregarded in the calculation of motor efficiency.

When the manufacturer offers a motor of the same design with and without a brake the test of motor efficiency may be done on a motor without the brake. The determined efficiency may then be used as the rating of both motor and brake motor.

NOTE When removing auxiliary devices for efficiency testing by third party labs the motor or auxiliaries may be permanently damaged.

4.2 Efficiency rating

The efficiency η_{90} of a motor shall be determined from the losses obtained according to IEC TS 60034-2-3 at 90 % of rated speed n_{90} and at rated torque T_N or at full-load torque T_{100} in relation to the output power at that operating point P_{90} :

$$P_{90} = 0,9 \cdot P_N \text{ and } n_{90} = 0,9 \cdot n_N$$

When a rated torque T_N is not provided by the manufacturer, the full-load torque T_{100} shall be determined from the rated speed and rated output power by applying the following formula:

$$T_{100} = \frac{P_N}{2\pi \cdot n_N}$$

with P_N in [W], n_N in [1/s], T_{100} in [Nm].

Thermal equilibrium of the motor shall be maintained at the beginning of the tests.

NOTE 1 Rated speed is usually the highest speed in the base speed-range.

NOTE 2 The requirement to test at 90 % of rated speed will ensure that the motor is operated at full magnetic flux (full voltage) regardless of the voltage drop in the internal electronic switches of the frequency converter. This procedure is harmonized with the requirements of frequency converter (CDM) testing included in IEC 61800-9-2.

NOTE 3 There is usually a slight drop of the motor efficiency when determined at 90 % rated speed and 90 % rated power compared to the efficiency determined at rated speed and rated power. This is covered by the

reduction of the nominal efficiency limits required for the IE classification according to this document when compared to the nominal efficiency limits given in IEC 60034-30-1.

The determined efficiency η_{90} shall be greater than or equal to the nominal efficiency η_n of the designated efficiency class (IE-code).

The nominal efficiency η_n of the IE1 - IE5 classes according to this document shall be calculated from the respective reference value η_{ref} given in Tables 2 to 6 or determined according to the interpolation given in 4.10, reduced by an allowance due to additional harmonic losses and efficiency determination at partial speed (90 %) by applying the following formula:

$$\eta_n = \frac{1}{1 + (1 + r_{HL}) \cdot \left(\frac{1}{\eta_{ref}} - 1 \right)}$$

The additional harmonic loss factor r_{HL} is:

$r_{HL} = 0,15$ (15 %) for motors with a rated output power up to and including 90 kW;

$r_{HL} = 0,25$ (25 %) for motors with a rated output power above 90 kW.

η_n and η_{ref} shall be given as numbers in the range of (0...1). The result shall be mathematically rounded to three decimal places or one decimal place when the efficiency is expressed as a percentage value.

NOTE 4 The change of the value of r_{HL} at 90 kW reflects the change of the switching frequency of the comparable converter as defined in IEC TS 60034-2-3.

4.3 Tolerances

Variations in materials, manufacturing processes and in testing result in motor-to-motor efficiency variations for a given motor design; the efficiency for a large population of motors of a single design is not a unique value but rather a band of efficiency. Therefore, the efficiency η_{90} of the population is to be evaluated from the determined efficiencies. The efficiency η_{90} of any motor shall not be less than the nominal efficiency η_n minus the tolerance in accordance with IEC 60034-1.

4.4 Classification

The designation of the energy-efficiency class consists of the letters “IE” (short for “International Energy-efficiency class”), directly followed by a numeral representing the classification according to Table 1.

Table 1 – IE-efficiency classification

Designation	Definition
IE1	Motors with an efficiency equal to or exceeding the reference values given in Table 2 diminished by the r_{HL} factor according to the procedure given in 4.2.
IE2	Motors with an efficiency equal to or exceeding the reference values given in Table 3 diminished by the r_{HL} factor according to the procedure given in 4.2.
IE3	Motors with an efficiency equal to or exceeding the reference values given in Table 4 diminished by the r_{HL} factor according to the procedure given in 4.2.
IE4	Motors with an efficiency equal to or exceeding the reference values given in Table 5 diminished by the r_{HL} factor according to the procedure given in 4.2.
IE5	Motors with an efficiency equal to or exceeding the reference values given in Table 6 diminished by the r_{HL} factor according to the procedure given in 4.2.

NOTE 1 Efficiency classes above IE5 are not planned.

NOTE 2 Table 1 is applicable to motors rated to any fundamental frequency.

To determine reference efficiency values with rated powers not given in Tables 2 to 6 within the range of 0,12 kW up to 200 kW, the interpolation formula and coefficients of 4.10 of this document can be applied.

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4.5 Nominal limits for efficiency class IE1

Table 2 – Reference values (%) for the calculation of IE1 nominal efficiency limits

Rated output power P_N [kW] at rated speed and full-load torque	Rated speed within 600 to 900 /min	Rated speed within 901 to 1 200 /min	Rated speed within 1 201 to 1 800 /min	Rated speed within 1 801 to 6 000 /min
0,12	31,0	38,3	50,0	45,0
0,18	38,0	45,5	57,0	52,8
0,20	39,7	47,6	58,5	54,6
0,25	43,4	52,1	61,5	58,2
0,37	49,7	59,7	66,0	63,9
0,40	50,9	61,1	66,8	64,9
0,55	56,1	65,8	70,0	69,0
0,75	61,2	70,0	72,1	72,1
1,1	66,5	72,9	75,0	75,0
1,5	70,2	75,2	77,2	77,2
2,2	74,2	77,7	79,7	79,7
3	77,0	79,7	81,5	81,5
4	79,2	81,4	83,1	83,1
5,5	81,4	83,1	84,7	84,7
7,5	83,1	84,7	86,0	86,0
11	85,0	86,4	87,6	87,6
15	86,2	87,7	88,7	88,7
18,5	86,9	88,6	89,3	89,3
22	87,4	89,2	89,9	89,9
30	88,3	90,2	90,7	90,7
37	88,8	90,8	91,2	91,2
45	89,2	91,4	91,7	91,7
55	89,7	91,9	92,1	92,1
75	90,3	92,6	92,7	92,7
90	90,7	92,9	93,0	93,0
110	91,1	93,3	93,3	93,3
132	91,5	93,5	93,5	93,5
160	91,9	93,8	93,8	93,8
200	92,5	94,0	94,0	94,0
250	92,5	94,0	94,0	94,0
315	92,5	94,0	94,0	94,0
355	92,5	94,0	94,0	94,0
400	92,5	94,0	94,0	94,0
450	92,5	94,0	94,0	94,0
500 up to 1 000	92,5	94,0	94,0	94,0

NOTE The tabled values are identical to the nominal values given for IE1 classification in IEC 60034-30-1.

4.6 Nominal limits for efficiency class IE2

Table 3 – Reference values (%) for the calculation of IE2 nominal efficiency limits

Rated output power P_N [kW] at rated speed and full-load torque	Rated speed within 600 to 900 /min	Rated speed within 901 to 1 200 /min	Rated speed within 1 201 to 1 800 /min	Rated speed within 1 801 to 6 000 /min
0,12	39,8	50,6	59,1	53,6
0,18	45,9	56,6	64,7	60,4
0,20	47,4	58,2	65,9	61,9
0,25	50,6	61,6	68,5	64,8
0,37	56,1	67,6	72,7	69,5
0,40	57,2	68,8	73,5	70,4
0,55	61,7	73,1	77,1	74,1
0,75	66,2	75,9	79,6	77,4
1,1	70,8	78,1	81,4	79,6
1,5	74,1	79,8	82,8	81,3
2,2	77,6	81,8	84,3	83,2
3	80,0	83,3	85,5	84,6
4	81,9	84,6	86,6	85,8
5,5	83,8	86,0	87,7	87,0
7,5	85,3	87,2	88,7	88,1
11	86,9	88,7	89,8	89,4
15	88,0	89,7	90,6	90,3
18,5	88,6	90,4	91,2	90,9
22	89,1	90,9	91,6	91,3
30	89,8	91,7	92,3	92,0
37	90,3	92,2	92,7	92,5
45	90,7	92,7	93,1	92,9
55	91,0	93,1	93,5	93,2
75	91,6	93,7	94,0	93,8
90	91,9	94,0	94,2	94,1
110	92,3	94,3	94,5	94,3
132	92,6	94,6	94,7	94,6
160	93,0	94,8	94,9	94,8
200 up to 1 000	93,5	95,0	95,1	95,0

NOTE The tabled values are identical to the nominal values given for IE2 classification in IEC 60034-30-1.

4.7 Nominal limits for efficiency class IE3

Table 4 – Reference values (%) for the calculation of IE3 nominal efficiency limits

Rated output power P_N [kW] at rated speed and full-load torque	Rated speed within 600 to 900 /min	Rated speed within 901 to 1 200 /min	Rated speed within 1 201 to 1 800 /min	Rated speed within 1 801 to 6 000 /min
0,12	50,7	57,7	64,8	60,8
0,18	58,7	63,9	69,9	65,9
0,20	60,6	65,4	71,1	67,2
0,25	64,1	68,6	73,5	69,7
0,37	69,3	73,5	77,3	73,8
0,40	70,1	74,4	78,0	74,6
0,55	73,0	77,2	80,8	77,8
0,75	75,0	78,9	82,5	80,7
1,1	77,7	81,0	84,1	82,7
1,5	79,7	82,5	85,3	84,2
2,2	81,9	84,3	86,7	85,9
3	83,5	85,6	87,7	87,1
4	84,8	86,8	88,6	88,1
5,5	86,2	88,0	89,6	89,2
7,5	87,3	89,1	90,4	90,1
11	88,6	90,3	91,4	91,2
15	89,6	91,2	92,1	91,9
18,5	90,1	91,7	92,6	92,4
22	90,6	92,2	93,0	92,7
30	91,3	92,9	93,6	93,3
37	91,8	93,3	93,9	93,7
45	92,2	93,7	94,2	94,0
55	92,5	94,1	94,6	94,3
75	93,1	94,6	95,0	94,7
90	93,4	94,9	95,2	95,0
110	93,7	95,1	95,4	95,2
132	94,0	95,4	95,6	95,4
160	94,3	95,6	95,8	95,6
200 up to 1 000	94,6	95,8	96,0	95,8

NOTE The tabled values are identical to the nominal values given for IE3 classification in IEC 60034-30-1.

4.8 Nominal limits for efficiency class IE4

Table 5 – Reference values (%) for the calculation of IE4 nominal efficiency limits

Rated output power P_N [kW] at rated speed and full-load torque	Rated speed within 600 to 900 /min	Rated speed within 901 to 1 200 /min	Rated speed within 1 201 to 1 800 /min	Rated speed within 1 801 to 6 000 /min
0,12	62,3	64,9	69,8	66,5
0,18	67,2	70,1	74,7	70,8
0,20	68,4	71,4	75,8	71,9
0,25	70,8	74,1	77,9	74,3
0,37	74,3	78,0	81,1	78,1
0,40	74,9	78,7	81,7	78,9
0,55	77,0	80,9	83,9	81,5
0,75	78,4	82,7	85,7	83,5
1,1	80,8	84,5	87,2	85,2
1,5	82,6	85,9	88,2	86,5
2,2	84,5	87,4	89,5	88,0
3	85,9	88,6	90,4	89,1
4	87,1	89,5	91,1	90,0
5,5	88,3	90,5	91,9	90,9
7,5	89,3	91,3	92,6	91,7
11	90,4	92,3	93,3	92,6
15	91,2	92,9	93,9	93,3
18,5	91,7	93,4	94,2	93,7
22	92,1	93,7	94,5	94,0
30	92,7	94,2	94,9	94,5
37	93,1	94,5	95,2	94,8
45	93,4	94,8	95,4	95,0
55	93,7	95,1	95,7	95,3
75	94,2	95,4	96,0	95,6
90	94,4	95,6	96,1	95,8
110	94,7	95,8	96,3	96,0
132	94,9	96,0	96,4	96,2
160	95,1	96,2	96,6	96,3
200	95,4	96,3	96,7	96,5
250	95,4	96,5	96,7	96,5
315 up to 1 000	95,4	96,6	96,7	96,5

NOTE The tabled values are identical to the nominal values given for IE4 classification in IEC 60034-30-1.

4.9 Nominal limits for efficiency class IE5

Table 6 – Reference values (%) for the calculation of IE5 nominal efficiency limits

Rated output power P_N [kW] at rated speed and full-load torque	Rated speed within 600 to 900 /min	Rated speed within 901 to 1 200 /min	Rated speed within 1 201 to 1 800 /min	Rated speed within 1 801 to 6 000 /min
0,12	67,4	69,8	74,3	71,4
0,18	71,9	74,6	78,7	75,2
0,20	73,0	75,7	79,6	76,2
0,25	75,2	78,1	81,5	78,3
0,37	78,4	81,6	84,3	81,7
0,40	78,9	82,2	84,8	82,3
0,55	80,6	84,2	86,7	84,6
0,75	82,0	85,7	88,2	86,3
1,1	84,0	87,2	89,5	87,8
1,5	85,5	88,4	90,4	88,9
2,2	87,2	89,7	91,4	90,2
3	88,4	90,6	92,1	91,1
4	89,4	91,4	92,8	91,8
5,5	90,4	92,2	93,4	92,6
7,5	91,3	92,9	94,0	93,3
11	92,2	93,7	94,6	94,0
15	92,9	94,3	95,1	94,5
18,5	93,3	94,6	95,3	94,9
22	93,6	94,9	95,5	95,1
30	94,1	95,3	95,9	95,5
37	94,4	95,6	96,1	95,8
45	94,7	95,8	96,3	96,0
55	94,9	96,0	96,5	96,2
75	95,3	96,3	96,7	96,5
90	95,5	96,5	96,9	96,6
110	95,7	96,6	97,0	96,8
132	95,9	96,8	97,1	96,9
160	96,1	96,9	97,2	97,0
200 up to 1 000	96,3	97,0	97,4	97,2

4.10 Interpolation of reference values at intermediate powers

To determine reference values for variable speed motors with rated powers not given in Tables 2, 3, 4, 5 and 6 within the range of 0,12 kW up to 200 kW the following formula shall be applied:

$$\eta_{\text{ref}} = A \cdot [\log_{10}\{P_N\}]^3 + B \cdot [\log_{10}\{P_N\}]^2 + C \cdot [\log_{10}\{P_N\}] + D$$

Where A , B , C , D are interpolation coefficients (see Table 7 and Table 8); P_N is given in [kW].

NOTE The formula and interpolation coefficients were mathematically derived to create a best fitting curve for the desired reference values. They do not have a physical meaning.

The resulting reference value (%) shall be rounded to the nearest tenth, i.e. xx,x %.

Table 7 – Interpolation coefficients for rated powers 0,12 kW up to 0,64 kW

IE-Code	Coefficients	Rated speed within 600 to 900 /min	Rated speed within 901 to 1 200 /min	Rated speed within 1 201 to 1 800 /min	Rated speed within 1 801 to 6 000 /min
IE1	A	5,9466	–45,9652	16,7271	11,924
	B	7,9458	–87,1474	12,7136	6,3699
	C	40,441	–8,2383	25,947	30,0509
	D	66,146	68,7303	76,174	76,6136
IE2	A	6,4855	–15,9218	17,2751	22,4864
	B	9,4748	–30,258	23,978	27,7603
	C	36,852	16,6861	35,5822	37,8091
	D	70,762	79,1838	84,9935	82,458
IE3	A	–0,5896	–17,361	7,6356	6,8532
	B	–25,526	–44,538	4,8236	6,2006
	C	4,2884	–3,0554	21,0903	25,1317
	D	75,831	79,1318	86,0998	84,0392
IE4	A	–4,9735	–13,0355	8,432	–8,8538
	B	–21,453	–36,9497	2,6888	–20,3352
	C	2,6653	–4,3621	14,6236	8,9002
	D	79,055	82,0009	87,6153	85,0641
IE5	A	–9,5776	–6,1120	11,0118	–7,0239
	B	–30,1627	–23,1331	8,3635	–16,9944
	C	–4,5962	1,6331	16,0368	8,1621
	D	81,2564	86,0990	90,5323	87,7915